FloodScope - Website User Guide

Research prototype - not a government or emergency service

Launch

Run START_WINDOWS.bat or install requirements and run python -m uvicorn backend.main:app --reload. Open http://127.0.0.1:8000.

Cover page

Choose Explore Location, Synthetic Demo, or Analyze Geographic Data.

Instant demonstration

Click Try Synthetic Demo. The polygon area and infrastructure intersection are computed from fictional GeoJSON, not real flooding.

Search

Search a location and choose the result. Change date and radius. Map backgrounds are not event-date satellite images.

Satellite

Find satellite scenes to discover Sentinel-1 and Sentinel-2 catalogue metadata, not classified flooding.

SAR change

Upload calibrated and co-registered before/after Sentinel-1 VV backscatter GeoTIFFs in dB; run radar-change candidate detection.

Infrastructure

Upload flood polygon and pre-event assets GeoJSON; calculate flooded area and exposed buildings, roads and bridges.

Cut-off villages

Upload roads, settlement/hospital points and flood polygon. Results are indicative, not verified road closures.

DEM

Upload Copernicus DEM GeoTIFF and trace from the selected map marker. This is not a hydraulic forecast.

Validation

Upload prediction and independent reference polygon GeoJSON; calculate IoU. EMSR927 is for validation only.

Image preview

Upload PNG/JPG for approximate map overlay; images are not automatically georeferenced.

Reports

Download assessment or analysis PDF after computing values. Hindi/Nepali report uses Latin transliteration.

Known gaps

No trained segmentation AI, automatic raw Sentinel preprocessing, real-event validation or verified financial losses. Not an emergency system.